Collective damping of two driven emitters

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Abstract. We study the Rabi oscillation of two emitters coupled by the dipole-dipole interaction. The decay rate sets the damping and the dressed states give the spectrum, a Mollow triplet with sidebands at twice the Rabi frequency. Equation (1) gives the master equation.

2. Model

The mixing angle theta obeys cot 2theta = Delta / 2 Omega (eq. 2). The secular approximation removes fast terms (eq. 3). The Bloch sphere picture helps.

Figure 1 shows the spectrum.

3. Results

The superradiance of the pair doubles the decay rate; subradiance halves it (eq. 7).

We thank the Dicke states for existing.